YEAR 11 MATHS METHODS INVESTIGATIVE TASK 3

Sequences

Prerequisites
Knowledge of:

- Arithmetic progressions
- Geometric progressions

Instructions

- Show all working for each part of each question.
- You may use your textbook and your notes.
- You may use your calculator but show sufficient working so your answer can be checked (especially if you use a spreadsheet) and your reasoning is clear.

Sequences

1 Consider the sequence 3, 7, 11, 15, …

Assuming the pattern continues, find:

a t_1

b d

c a formula for the nth term

d the 25th term

e a formula for the sum of the first n terms and hence the sum of the first 100 terms in the sequence.

[1 + 1 + 2 + 1 + 2 = 7 marks]

2 A certain sequence begins with 3 and every term after the first is double the previous term.

a Write down the first four terms of this sequence.

b What kind of sequence is this?

c Write down a formula for the nth term.

d Use a suitable formula to find the sum of the first 12 terms.

[1 + 1 + 2 + 2 = 6 marks]

3 The first term of an arithmetic series is −5 and the last term is 2005. Find the sum of the series if there are 400 terms in the series.

[3 marks]

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4 Without using a formula, use first principles to find a value for

a The total of all the odd numbers from 1 to 999.

b The total of all the odd numbers from 1 to n.

[4 + 5 = 9 marks]

5 Consider the series $1 + x + x^2 + x^3 + x^4 + \cdots + x^n$.

a Describe the series in terms of the two significant parameters.

b Hence write an expression for a general term t_n.

c How many terms are there in this series?

d Write an expression for the sum of the series.

e Hence find an expression for $1 + x + x^2 + x^3 + x^4 + \cdots + x^{19}$.

f Hence factorise $x^{20} - 1$.

g Extend this to a general rule for factorising $x^n - 1$. Prove your answer from first principles. (*Hint*: First find a rule for $S_n = 1 + x + x^2 + x^3 + \cdots + x^{n-1}$.)

[2 + 2 + 1 + 2 + 1 + 1 + 4 = 13 marks]

6 After her grandson, Ben, planted a new shrub in her garden, Mrs Green became concerned that the shrub would grow to block out the light to her window. She measured the height of the shrub carefully each month for several months. The house was slightly higher than ground level at this point and the window ledge was 1.7 metres above the ground.

The measurements of the shrub are as follows.

Month	1	2	3	4
Height at start of month (m)	0.7	0.9	1.06	1.188
Growth during that month (cm)	20	16	12.8	10.24

a Describe fully the pattern for growth in the table.

b How many centimeters would the plant grow

i during the 6th month

ii during the nth month

c What would the height of the plant be at the end of

i 6 months?

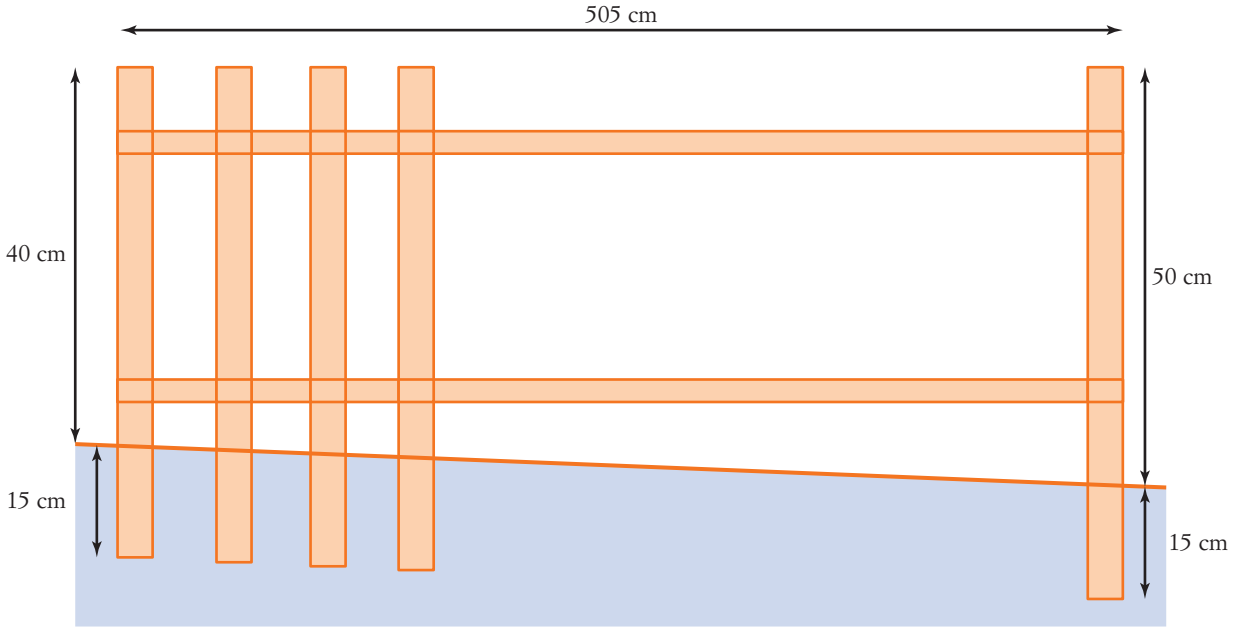
ii n months?

d Assuming the plant continued to grow at the same rate, would the plant ever grow too tall and block the light from the Mrs Green's window?

[3 + 2 + 2 + 1 = 8 marks]

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7 Ben also decided to replace the front of a little fence that kept the dog out of his grandmother's vegetable patch. The fence was on a slightly sloping ground. Fortunately for Ben, the slope was uniform. At one end the fence post was 40 cm above the ground. At the other end the fence post was 50 cm above the ground. All fence posts are 15 cm below the ground for stability. Fence posts are 5 cm wide and 20 cm apart. The total distance from the first post to the last is 505 cm. There are two longer pieces of wood, the same as the posts along the length of the fence, i.e. 5 cm wide.



The fence posts and the pieces of wood along the length of the fence are all going to be cut from lengths of wood bought at the local hardware store. The lengths of wood are available in 5 cm widths.

a How many posts will Ben need to cut?

b Will the lengths of the posts form an arithmetic progression, a geometric progression, or some other kind of sequence? Explain your answer.

c How long will the first post be?

d How long will the last post be?

e Using a suitable formula, calculate how much longer each post will be than the one before it (i.e. on its left).

f Using a suitable formula, calculate the total length of wood Ben will need. Don't forget about the wood going across.

The lengths of wood can be 3 m, 3.6 m, 4.8 m or 5.1 m long.

g What wood lengths would it be best for Ben to buy? Explain fully.

[2 + 2 + 1 + 1 + 4 + 4 + 10 = 24 marks]

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